

AI Engineer

Donya Elshahawy

I'm a fourth-year **Artificial Intelligence Engineering** student who combines technical precision with real curiosity about how machines learn, understand, and make decisions. I work hands-on across **Machine Learning**, **Deep Learning**, **Natural Language Processing**, **Computer Vision**, and **AI Agents** — turning ideas into solutions that actually run, using Python, C, C++, Java, and SQL.

I bring logic to the table and leave room for lightness — because the best solutions don't have to be complicated to be serious, and great work is always better with curiosity and a little bit of energy behind it.

ABOUT ME

About Me

I help businesses and individuals turn their data and ideas into practical AI solutions that automate tasks, uncover valuable insights, and solve real-world problems using Machine Learning, Deep Learning, and Data Analysis.

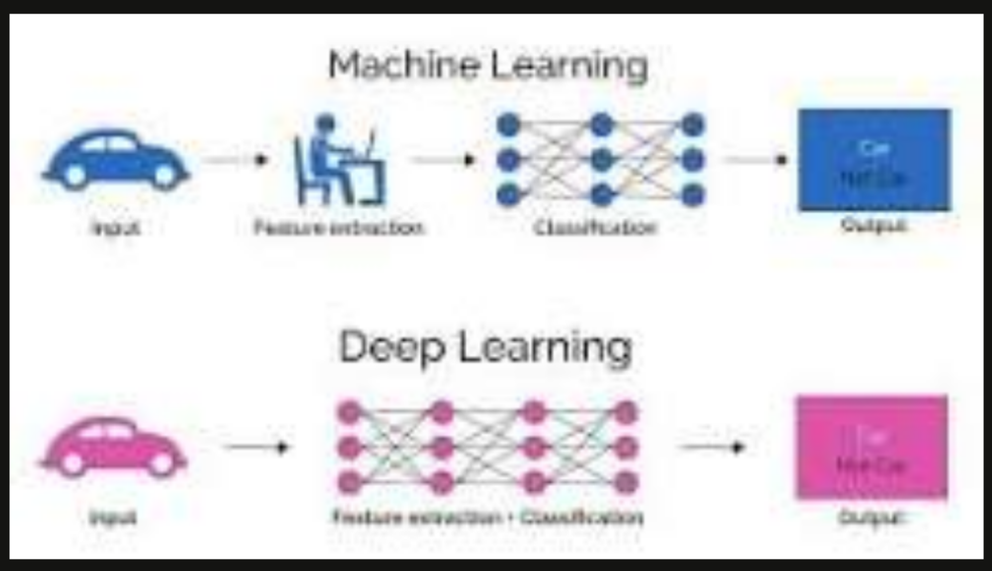
I'm a fourth-year Artificial Intelligence Engineering student at New Ismailia National University and an AI & Data Science enthusiast, passionate about building practical, intelligent solutions using Machine Learning, Deep Learning, Data Analysis, Computer Vision, and Natural Language Processing.

Through academic projects and hands-on technical training at NTI and DEPI, I've developed, evaluated, and deployed AI models on real-world datasets — using Python, Scikit-learn, Pandas, TensorFlow, Keras, and PyTorch. I've worked on multiple Machine Learning and Deep Learning projects end to end, from data collection and preprocessing to evaluation and deployment.

What sets me apart is range: I don't stay in one lane. My work spans CNN-based medical image classification, full relational database systems, embedded microcontroller projects, rule-based expert systems, and swarm-optimization scheduling — all built with the same care for clean code, clear evaluation, and honest documentation. I'm actively seeking opportunities to grow as an AI Engineer, contribute to impactful projects, and keep developing my technical, problem-solving, teamwork, and communication skills.

I'm fluent in **Arabic (native)** and **English (proficient)**, and I genuinely enjoy the collaborative side of engineering — explaining a model's reasoning to a non-technical teammate is as much a part of the job to me as tuning its hyperparameters.





EDUCATION

Education

2023
—
2028

B.Sc. in Artificial Intelligence Engineering

New Ismailia National University — Ismailia, Egypt

Fourth-year student building solid foundations in machine learning, deep learning, data structures, and intelligent systems. The program blends theory with hands-on practice across Python, C, C++, Java, and SQL, with a particular focus on Natural Language Processing and Computer Vision — and how AI can be applied to solve meaningful, real-world challenges.

Grade: Excellent

TRAINING & PROGRAMS

DEPI — Digital Egypt Pioneers Initiative

AI & Data Science Trainee — Microsoft ML Engineer Track

Practical AI/ML solutions in Python: data preprocessing, visualization, model development and evaluation, plus MLOps concepts using Azure AI, MLflow, and Hugging Face.

2026 — Present

NTI — National Telecommunication Institute

Machine Learning

Core ML training covering data preprocessing, predictive modeling, and classic ML algorithms.

07/2025 — 08/2025

NTI — National Telecommunication Institute

AI for Business

Applying machine learning and CNNs to real business use cases and decision-making problems.

07/2026 — 08/2026

IEEE — Suez Canal University Student Branch

AI Training

Foundational AI concepts and an introduction to AI agents.

03/2026

SKILLS

Skills



Programming Languages

Python, C, C++, Java, SQL

AI & Machine Learning

Machine Learning, Deep Learning, CNN, RNN, LSTM

NLP, Computer Vision, Transfer Learning

Generative AI, AI Agents, Clustering

Data & Modeling

Data Analysis, Data Preprocessing

Feature Engineering, Model Evaluation

Model Deployment, MLOps

Languages & Libraries

TensorFlow, Keras, PyTorch, Scikit-learn

Pandas, NumPy, Matplotlib

Tools & Platforms

Git, GitHub, Kaggle, Google Colab

Microsoft Azure, Streamlit, MLflow, Hugging Face

Soft Skills

Leadership, Teamwork, Effective Communication

Problem Solving, Critical & Analytical Thinking

Time Management, Creativity, Continuous Learning

Work Experience

AI & Data Science Trainee

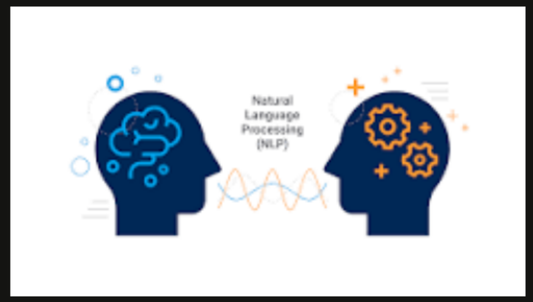
2026 — Present

Digital Egypt Pioneers Initiative (DEPI) — Microsoft Machine Learning Engineer Track

- Developed practical AI and Machine Learning solutions using Python, covering data preprocessing, visualization, model development, and evaluation.
- Applied Machine Learning and Deep Learning techniques to practical projects involving NLP, Computer Vision, and predictive modeling.
- Practiced AI model development, evaluation, deployment, and MLOps concepts using Azure AI, MLflow, and Hugging Face.
- Worked on practical AI projects while developing problem-solving, critical thinking, teamwork, and communication skills.
- Collaborated in a structured, track-based learning environment designed around real Microsoft ML engineering practices.
- Gained hands-on experience building end-to-end ML pipelines on real-world datasets, from raw data to a deployable model.
- Applied algorithms including Logistic Regression, SVM, Decision Trees, and Neural Networks, with feature engineering and evaluation done in Pandas and Scikit-learn.

OFFERED SERVICES

Offered Services



01

Machine Learning & Deep Learning Model Development

Design, train, and evaluate CNN and transfer-learning models (e.g. EfficientNetB0) for classification and predictive tasks, from data prep to final evaluation.

02

Computer Vision Solutions

Medical and industrial image classification systems — such as cancer tissue classification and building crack detection — using TensorFlow, Keras, and OpenCV.

03

NLP & Speech Processing

Speech transcription and translation systems built on OpenAI Whisper, with frequency-domain analysis via FFT and FIR/IIR filtering.

04

Data Cleaning, Analysis & Visualization

Cleaning, exploring (EDA), and visualizing data with Pandas and Matplotlib to uncover patterns and support real decisions.

05

Desktop & GUI Applications

Python and Tkinter tools — from ensemble ML model comparison dashboards to explainable, rule-based expert systems.

06

Database & C++ System Design

Full relational database design (SQL Server) and OOP-based management systems built with structured C++ data structures.

07

Embedded Systems & Automation

Embedded C and Arduino projects — from traffic-light state-machine controllers to real-time gas-leak detection systems.

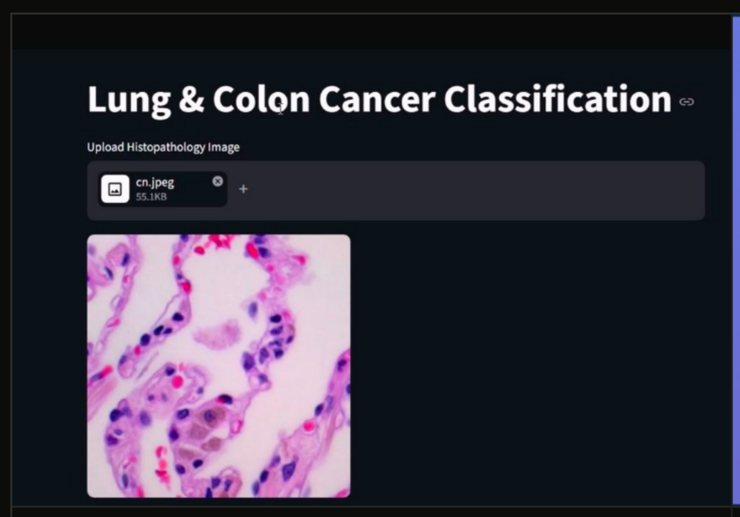
08

AI Consulting & Mentorship

Guidance and mini-projects for AI students and early learners looking for a practical, hands-on start in the field.

SELECTED PROJECTS

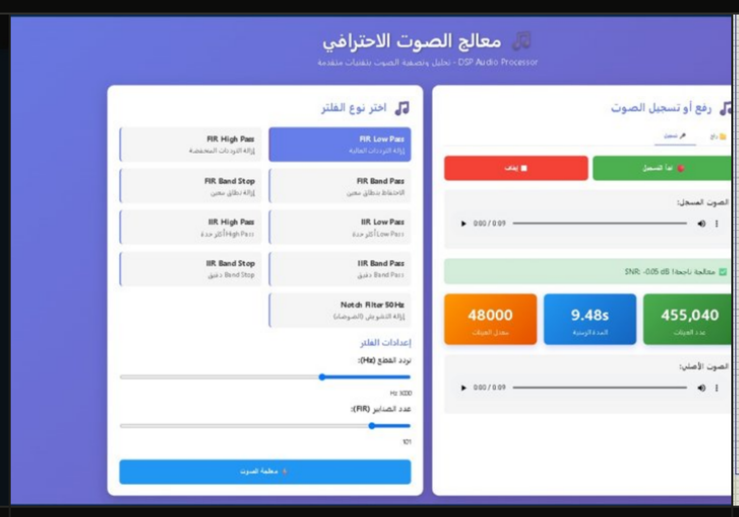
Projects



Cancer Classification using Deep Learning

Python · TensorFlow · Keras · CNN · EfficientNetB0

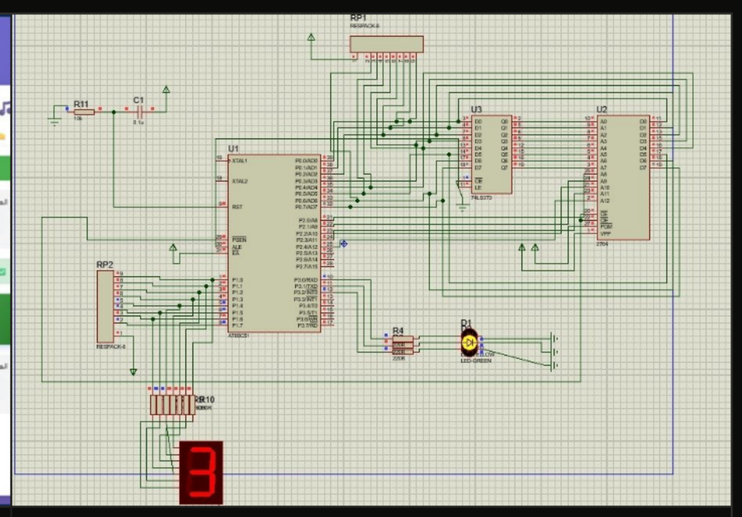
A CNN-based image classification model to classify lung and colon tissue images, with preprocessing, augmentation, and fine-tuned EfficientNetB0 via transfer learning.



DSP Audio Processing & Speech Analysis

Python · DSP · FFT · FIR/IIR · Whisper

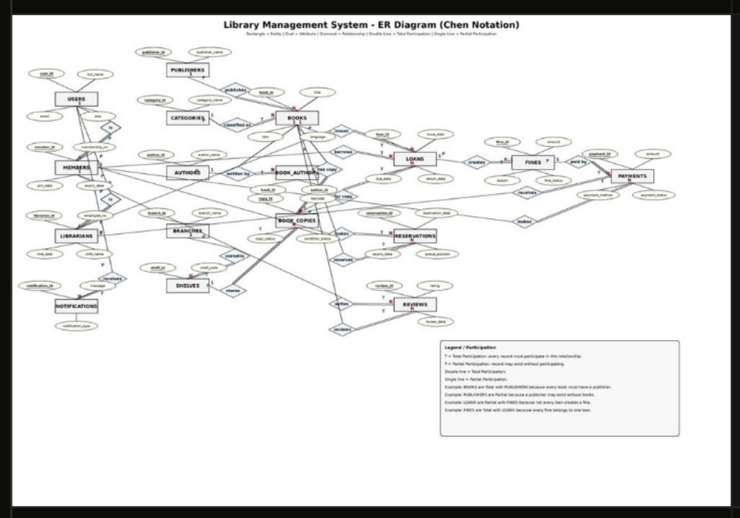
An interactive audio system with time- and frequency-domain analysis and FIR/IIR filtering, integrated with Whisper for transcription and translation.



Traffic Light Controller with 7-Segment Countdown

Embedded C · AT89C51 · Proteus · Keil µVision

A simulated AT89C51 traffic-light controller with state-machine logic and a 7-segment display for real-time countdowns.



Library Management System

SQL Server · Database Design

A relational database covering books, members, loans, reservations, fines, and payments, with automated issuing and overdue detection.

Ensemble AI Model Training GUI

Python · Scikit-learn · XGBoost · Tkinter

A GUI for training and comparing Decision Tree, Bagging, AdaBoost, Gradient Boosting, XGBoost, and Stacking models with built-in evaluation.

Laptop Selection Expert System

Python · Experta · Tkinter

A rule-based expert system using production rules, salience, and Certainty Factor to recommend laptops, with rule tracing for transparent decisions.

Building Crack Detection

TensorFlow · Keras · OpenCV · FFNN/RNN-GRU/CNN

An image classifier distinguishing cracked from non-cracked surfaces, achieving 91.54% validation accuracy with class weighting.

Optimized Weekly Study Scheduler Using PSO

Python · NumPy · Pandas · Matplotlib

A weekly study scheduler using Particle Swarm Optimization with a priority model based on difficulty, importance, and exam proximity.

Gas Detection System

Arduino · C/C++ · Gas Sensor

An Arduino-based gas detection system that triggers a real-time alert when gas levels exceed a set threshold.

ACHIEVEMENTS & CERTIFICATIONS

Achievements & Certifications



AI for Business

ITIDA · NTI

07/2026 – 08/2026



Overview of AI

Huawei ICT Academy

10/2026



AI Track — Tech Nights

IEEE Suez Canal University Student Branch

03/2026



Generative AI

Mahara-Tech

In Progress



Getting Started with Deep Learning

NVIDIA

08/2025



Machine Learning

ITIDA · NTI

07/2025 – 08/2025



Java and SQL Summer Training

Suez Canal Authority

08/2025



Intro to Programming

Kaggle

08/2025



Natural Language Processing (NLP)

DataCamp

Completed



ISEF Certificate of Appreciation

Technology Development Center

2018 – 2019

CONTACT ME

Contact Me

I'm always open to new opportunities, collaborations, or just a friendly chat about AI and creativity!

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LINKEDIN	linkedin.com/in/donya-elshahawey-a76aa031

Feel free to reach out — let's create something impactful together!

Thank You

Thank you for taking the time to explore my portfolio! I'm truly grateful for your interest in my work and journey as an AI Engineer.

If you believe in the power of innovation, creativity, and collaboration — let's connect and create something amazing together.

Your visit means a lot. Until we meet or work together — stay curious, stay inspired!

Donya Elshahawy

